AI 201 @ Workplace 2026

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A (brave?) new world...

In a world of infinite limitless plausible-looking text and code, there are two options...

👎 We drown in “slop” (*low-quality digital content, often produced in large quantities by artificial intelligence, that lacks depth or value*)

👍 We provide:

- Direction
- Judgement
- Taste
- Values
- Relevance
- Curation
- Implementation
- Alignment
- Specification (or Intent)
- Wisdom (or embodied taste)

Where We've Been vs. Where We're Going (2026)

AI 101 (The Basics)	AI 201 (Today)
What is an LLM	How do LLMs "think"
How to write a basic prompt	How to engineer prompt systems
Getting answers	Getting *reliable* answers
One-shot questions	Multi-step workflows
"Help me with x"	"Here's my process for X"

We aren't trying to use AI just to do more, we're learning to use AI to do more if it's needed and possible, to do better if it gives us a meaningful edge, and to free cognitive space for curiosity, exploration, and the work that actually energizes us. The goal (as I interpret it and advocate for) is not maximum productivity, but to work in ways that are sustainable, effective, and maybe even enjoyable.

What we'll cover today

1. **How LLMs Actually Work** (The mental model that changes everything)
2. **The 6-Element Prompt Framework** (Structure that works every time)
3. **Advanced Techniques** (Self-correction, meta-prompting, verification)
4. **Workflow Design** (From single prompts to systems)
5. **Task Decomposition** (Breaking complex work into AI-friendly pieces)
6. **Quality Control** (How to trust AI outputs)
7. **Practical Applications** (Your actual work scenarios and available toolset)

Don't worry about memorizing anything, all materials and examples will be available in the description below. Focus on following the logic and understanding the *hows* and the *whys*

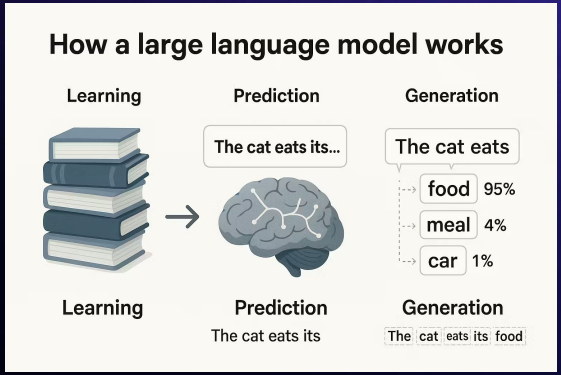
1. How LLMs Actually Work: The Mental Model Shift

Wrong mental model: LLMs are a search engine that retrieves stored information

Right mental model: LLMs are a pattern-completion engine that generates probable continuations

Why this matters:

- LLMs don't "know" things; they predict what comes next
- Quality of output depends on how you set up the input pattern
- You're not asking a question; you're providing a template to complete



1. How LLMs Actually Work: Implications for your prompts

If LLMs are a pattern-completer...	Then you should...
They continue patterns	Provide clear patterns to continue
They follow context cues	Give explicit context and role
They aim for "most likely"	Constrain the possibility space
They can be inconsistent	Build in verification steps

Key insight: The more structure you provide, the less the LLM has to "guess", and guessing is where errors happen.

1. How LLMs Actually Work: The Parameters You Control

Understand what these do conceptually:

- **Temperature/Creativity**: Low = focused and consistent; High = varied and exploratory
- **Context Window**: How much information the LLM can "see" at once
- **Instruction Following**: Modern models are trained to follow structured instructions

Practical tip: When you want consistency (reports, analysis), be more specific and constrained. When you want ideas (brainstorming), be more open-ended.

2. The 6-Element Prompt Framework:

Anatomy of a Professional Prompt

Every effective prompt can include up to 6 elements:

- Role – Who is the LLM in this interaction?
- Goal – What specific outcome do you need?
- Context – What background information matters?
- Format – How should the output be structured?
- Examples – What does "good" look like?
- Constraints – What should be avoided or limited?

You don't always need all 6, but knowing all 6 helps you diagnose when prompts fail.

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 1 - Role

What it does: Sets the LLM's perspective, vocabulary, and approach

Examples:

- "You are a senior project manager reviewing this proposal..."
- "You are a technical writer creating documentation for non-experts..."
- "You are a financial analyst identifying risks in this report..."

Why it works: The LLM adjusts its language, depth, and priorities based on the role

Common mistake: Skipping the role and getting generic, unfocused responses

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 2 - Goal

What it does: Defines success clearly

Weak goal: "Help me with this presentation"
Strong goal: "Create a 5-slide outline for a client presentation that explains our Q3 results, emphasizing the 15% efficiency improvement"

The test: Could someone else verify if the output achieved the goal?

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 3 - Context

What it does: Provides the information the LLM needs to work with

Types of context:

- **Background info:** "This client has been with us for 3 years and recently expanded operations..."
- **Document content:** Attach or paste relevant text
- **Situation:** "We're responding to a complaint about delayed delivery..."
- **Constraints from reality:** "Budget is €50,000 and timeline is 6 weeks..."

Rule: If a smart colleague would need this info to help you, so does the LLM.

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 4 - Format

What it does: Specifies exactly how you want the output structured

Examples:

- "Respond in bullet points, maximum 5 items"
- "Create a table with columns: Issue, Impact, Recommendation"
- "Write in email format with subject line and formal greeting"
- "Provide your answer as: Summary (2 sentences), Details (3-5 bullets), Next Steps (numbered list)"

Why it matters: You'll spend less time reformatting and can use outputs immediately.

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 5 - Examples

What it does: Shows the LLM exactly what "good" looks like

When to use:

- When format is unusual or specific
- When tone/style matters
- When you've had trouble getting the right output

Power move: Use one of your own best outputs as an example to maintain consistency.

2. The 6-Element Prompt Framework

Anatomy of a Professional Prompt

Element 6 - Constraints

What it does: Tells the LLM what NOT to do

Examples:

- "Do not include pricing information"
- "Keep the response under 200 words"
- "Avoid technical jargon—the audience is non-technical"
- "Do not recommend options outside our current service offerings"
- "If information is missing, ask for clarification rather than assuming"

Key insight: LLMs often over-deliver or go off-track. Constraints keep it focused.

2. The 6-Element Prompt Framework

Putting it all together: Template

```
**Role:** You are a [specific role] with expertise in [relevant domain].

**Goal:** [Clear, measurable outcome]

**Context:** [Background information, relevant documents, situation details]

**Format:** [Exactly how to structure the response]

**Examples:** (if needed)

[Sample input] → [Sample output]

**Constraints:**

- [What to avoid]

- [Limitations]

- [Style requirements]
```

3. Advanced Techniques

Why Basic Prompts Sometimes Fail

Even well-structured prompts can produce:

- **Hallucinations/Confabulations** – Confident but wrong information
- **Inconsistencies** – Contradictions within the same response
- **Drift** – Starting focused but wandering off-topic
- **Superficiality** – Technically correct but missing depth

Solution: Layer in techniques that make AI check its own work.

3. Advanced Techniques

Technique 1 - Chain-of-Verification (CoVe)

What it is: Force the LLM to generate, critique, and revise before giving final answer

How to use it:

```
[Your main request]

After providing your initial analysis:

1. List 3 ways your analysis could be incomplete or wrong

2. For each, check against the provided information

3. Provide a revised analysis addressing any issues found
```

When to use: Contract review, financial analysis, technical documents, anything where mistakes are **costly**

3. Advanced Techniques

Technique 2 - Self-Review Prompt

What it is: Ask the LLM to evaluate its own output against your criteria

Example addition to any prompt:

```
Before finalizing, check your response against these criteria:

- Did you address all parts of the question?

- Is the tone appropriate for the audience?

- Is anything unclear or could be misinterpreted?

- Are there any claims that need verification?

If any issues, revise accordingly.
```

3. Advanced Techniques

Technique 3 - Structured Thinking (for Complex Problems)

What it is: Force step-by-step reasoning through structure

Instead of: "What should we do about this problem?"

Use:

```
Analyze this problem using the following structure:

Step 1 - Define the core issue:

Step 2 - Identify key stakeholders affected:

Step 3 - List possible approaches (minimum 3):

Step 4 - Evaluate pros/cons of each:

Step 5 - Recommend an approach with reasoning:
```

Why it works: The structure forces sequential thinking rather than jumping to conclusions.

3. Advanced Techniques

Technique 4 - Multi-Perspective Analysis

What it is: Have the LLM consider multiple viewpoints before concluding

Example:

```
Analyze this proposal from three perspectives:

1. **Financial perspective:** Focus on costs, ROI, budget impact

2. **Operational perspective:** Focus on feasibility, resources, timeline

3. **Risk perspective:** Focus on what could go wrong

After analyzing all three, provide a synthesized recommendation that addresses all
perspectives.
```

When to use: Complex decisions, stakeholder communications, strategic planning

3. Advanced Techniques

Technique 5 - The "Prove It" Prompt

What it is: Ask the LLM to provide evidence for its claims

Add to any analytical prompt:

```
For each recommendation or conclusion, cite the specific information from the
provided documents that supports it. If you cannot cite specific support, note that
the point is based on general knowledge rather than the documents.
```

Why it works: Reduces hallucination by forcing the LLM to ground claims in provided content.

3. Advanced Techniques: Meta-Prompting

Definition: Using LLMs to improve how you use LLMs

The insight: The LLM has been trained on thousands of examples of effective prompting. You can ask it to help you prompt better.

Three applications:

1. Reverse prompting (LLM writes the prompt)
2. Prompt optimization (LLM improves your prompt)
3. Prompt debugging (LLM diagnoses why a prompt failed)

3. Advanced Techniques: Meta-Prompting

Reverse Prompting

When you don't know how to start:

```
You are an expert prompt engineer.

I need to [describe your goal].

Write the single most effective prompt that would make an AI produce excellent results for
this task. Consider:

- What context would be essential?

- What format would be most useful?

- What constraints would prevent common errors?

Then execute that prompt.
```

Best for: Unfamiliar tasks, when you're stuck, exploring what's possible

3. Advanced Techniques: Meta-Prompting

Prompt Optimization

When you have a prompt that's "okay" but not great:

```
Here's a prompt I've been using:

"[paste your current prompt]"

The outputs are [describe the issue: too vague, wrong format, missing depth, etc.]

Improve this prompt by:

1. Identifying what's missing or unclear

2. Adding necessary structure or constraints

3. Rewriting it for better results

Provide the improved version.
```

3. Advanced Techniques: Meta-Prompting

Prompt Debugging

When a prompt keeps failing:

```
I used this prompt:

"[paste your prompt]"

And got this output:

"[paste the problematic output]"

I expected: [describe what you wanted]

Diagnose why the prompt didn't work as intended and suggest specific fixes.
```

3. Advanced Techniques: Meta-Prompting

The Clarification Technique

When you're not sure what information the LLM needs:

```
I need help with [general description of task].

Before we begin:

1. What additional information would help you provide the best response?

2. Are there any ambiguities in my request you'd like clarified?

3. What assumptions are you making that I should confirm?

Please ask me these questions before proceeding.
```

Why it works: Surfaces gaps in your prompt before you get a bad output.

4. Workflow Design

From Single Prompts to Prompt Chains

Single prompt thinking: "Generate the report"

Workflow thinking (different prompts after each response):

1. Extract key data from these documents
2. Identify the three most important trends
3. Draft executive summary focused on those trends
4. Review draft for accuracy against source documents
5. Format for final presentation

Key shift: Complex tasks should be broken into steps, not crammed into one prompt or one conversation window.

4. Workflow Design

Chaining Pattern 1 - The Outline-Then-Execute

Step 1: Generate a structured outline

```
Create a detailed outline for [your task].

Include: main sections, key points for each, and what information/analysis each section needs.

Don't write the full content yet—just the outline.
```

Step 2: Review and adjust the outline yourself

Step 3: Execute section by section

```
Using this approved outline: [paste outline]

Write Section 2: [section title]

Include: [specific details for that section]
```

Why it works: You control the structure; LLM handles the writing.

4. Workflow Design

Chaining Pattern 2 - Extract-Analyze-Synthesize

Step 1: Extract

```
From the attached documents, extract all information related to [specific topic].

List each point with its source document.
```

Step 2: Analyze

```
Based on this extracted information: [paste extraction results]

Analyze for:

- Key patterns or trends

- Contradictions or gaps

- Implications for [your purpose]
```

Step 3: Synthesize

```
Using this analysis: [paste analysis]

Write a [document type] that [specific goal].
```

4. Workflow Design

Chaining Pattern 3 - Draft-Review-Revise

Step 1: Generate draft

```
[Your standard prompt for the task]
```

Step 2: AI self-review

```
Review this draft against these criteria:

- Accuracy: Are all facts verifiable?

- Completeness: Are all required elements present?

- Clarity: Would the intended audience understand easily?

- Tone: Does it match the required style?

Identify any issues.
```

Step 3: Revise

```
Based on this review: [paste review]

Revise the draft to address each identified issue.
```

4. Workflow Design

Building Your Workflow Library

Over time, develop reusable workflows for recurring tasks:

Task Type	Workflow Pattern
Document summarization	Extract → Verify key points → Format summary
Report writing	Outline → Section by section → Review → Polish
Email responses	Analyze incoming → Draft response → Tone check
Meeting prep	Extract from docs → Generate key points → Anticipate questions
Proposal review	Extract claims → Verify against data → Identify gaps → Draft feedback

5. Task Decomposition

The Manager's Mindset

"Delegate effectively. Break a task into pieces, just as you would for a capable but inexperienced colleague."

LLMs struggle with:

- Tasks requiring multiple types of thinking
- Implicit dependencies between steps
- Tasks where context from step 1 is needed in step 5

5. Task Decomposition

Signs You Need to Decompose

- Output only addresses part of your request
- LLM seems to lose track midway through
- Results are superficial despite detailed prompts
- You keep having to ask follow-up questions
- The task requires >2-3 distinct cognitive activities

5. Task Decomposition

Decomposition Framework

For any complex task, ask:

1. **What are the distinct activities?** (research, analyze, calculate, write, review)
2. **What dependencies exist?** (Step 3 needs output from Step 1)
3. **Where do I need to verify before proceeding?** (Critical decision points)
4. **What can run in parallel vs. what's sequential?**

Then design prompts for each distinct activity.

5. Task Decomposition

Example Decomposition

Original task: "Help me prepare for the quarterly business review meeting"

Step	Activity	Prompt Focus
1	Extract	Pull key metrics from Q3 reports
2	Compare	Benchmark against Q2 and targets
3	Identify	Find 3-5 key talking points (wins + concerns)
4	Anticipate	Generate likely questions from leadership
5	Prepare	Draft responses to anticipated questions
6	Format	Create meeting agenda with timing

Each step can be a separate prompt, with you reviewing between steps.

5. Task Decomposition

The "Least-to-Most" Approach

For problems that build on themselves:

Start with the simplest sub-problem, solve it, use that solution in the next problem.

Example for a complex analysis:

1. "First, just list the key data points in this document"
2. "Using those data points, identify the three main trends"
3. "For each trend, analyze causes and implications"
4. "Based on all the above, what are the strategic recommendations?"

Each step's output becomes context for the next.

6. Quality Control

The Trust Problem

LLM outputs can be:

- **Confidently wrong** (hallucinations stated as fact)
- **Subtly incomplete** (missing key considerations)
- **Contextually inappropriate** (technically correct but wrong for situation)

Your job: Build verification into your process, not after it.

6. Quality Control

Quality Control Layer 1 - Built-In Verification

Include verification instructions in the prompt itself:

```
[Your main task]

Quality checks before finalizing:

☐ All claims are supported by the provided documents

☐ No information has been assumed that wasn't given

☐ The format matches the specified requirements

☐ Tone is appropriate for [audience]

If any check fails, note it and explain what additional information would be
needed.
```

6. Quality Control

Quality Control Layer 2 - Self-Consistency Check

For important outputs, have the AI check its own work from a different angle:

```
You just produced this output: [paste output]

Now, as a critical reviewer, identify:

1. Any claims that aren't directly supported by the source material

2. Any logical gaps or assumptions

3. Any potential misinterpretations of the original request

4. Anything a skeptical reader might question

Be thorough—it's better to over-flag than to miss issues.
```

(you can also use different chat instances or even different LLM models for this!)

6. Quality Control

Quality Control Layer 3 - Explicit Uncertainty

Train yourself to ask for confidence signals:

```
[Your request]

For each major conclusion or recommendation:

- Rate your confidence (High/Medium/Low)

- If Medium or Low, explain what information would increase confidence

- If you're uncertain about anything, say so explicitly rather than guessing
```

6. Quality Control

Quality Control Layer 4 - Source Grounding

When working with documents:

```
[Your analysis request]

Important: For each point in your response, indicate which source document(s)
it comes from. Use this format:

- [Your point] [Source: Document name, relevant section]

If a point is based on general knowledge rather than the provided documents,
note: [Source: General knowledge]
```

6. Quality Control

The Human Verification Checklist

Always verify LLM outputs yourself for:

Check	Question to Ask
Factual accuracy	Can I verify these specific claims?
Completeness	Did it address everything I asked?
Appropriateness	Would I be comfortable with a colleague seeing this?
Context fit	Does this make sense for our specific situation?
Actionability	Can this actually be used as-is or does it need work?

6. Quality Control

Remember! As humans we uniquely provide:

- Direction
- Judgement
- Taste
- Values
- Relevance
- Curation
- Implementation
- Alignment
- Specification (or Intent)
- Wisdom (or embodied taste)

This is our specialty now!

7. Practical Applications

Prompts, Workflows, Templates in the description below!

Some of the examples included:

- Prompt Engineering & Optimization
- Document Analysis & Evaluation
- Meeting & Task Management
- Strategic Analysis & Organizational Insight
- Reporting & Proposal Generation
- Process & Incident Documentation
- Self-Assessment & Learning
- Etc...

7. Practical Applications

How to apply this to your work?

***About tedious work*:**

- Which tasks am I reluctant to do or feel low-energy about?
- Where do I write similar things repeatedly?
- What involves a lot of "translating" or reformatting?

***About bottlenecks*:**

- What slows me down most on a typical day?
- Where do I get stuck synthesizing information from multiple sources?
- What takes longer than it "should"?

***About quality*:**

- What would I do better with more time to think?
- Where would a "second opinion" help?

7. Practical Applications

1. **Pick ONE task** from your reflection, something you do regularly

2. **Describe it**: What inputs do you have? What output do you need? What's tedious about it?

3. **Design a prompt** using today's framework

4. **Evaluate honestly**:

    - Did it save time? Energy?
    - Did quality improve?
    - Was it worth the setup effort?

5. **Share what works**!

Not every task is a good fit for LLMs. That's useful information too!

7. Practical Applications

Final Thought

> ***"Prompt engineering is treating the LLM like a capable but literal-minded colleague who needs clear instructions, relevant context, and explicit expectations."***

The goal isn't to learn magic words, but to develop the habit of thinking clearly about what you need and communicating it precisely.

Your LLM is only as good as your ability to direct it.

The Past, Present, and (Near) Future...

Most are still here!

When	Stage	What It Means	What You Can Do
Early 2023	Basic Chatbots	AI answers questions from memory	Ask questions, get text answers (often wrong)
Mid 2023	Document Attachments	AI can read your files	"Summarize this PDF" or "Analyze this spreadsheet"
Late 2023	Custom Bots + Knowledge Bases	AI trained on your specific docs	Build a "company expert" that knows internal policies
2024	Tool-Using Agents	AI can take actions, not just talk	Autonomously send emails, query databases, call APIs
Early 2025	Looping Agents	AI works autonomously in cycles	"Build this feature" → AI codes, tests, fixes, repeats until it's done, navigating obstacles
Late 2025–2026	Orchestrators & Swarms	Multiple specialized AIs coordinate	Teams of agents: one plans, one researches, one executes

AI as a **tool**

AI as a **colleague**